STUDYNEST | CLASS 11 PHYSICS

QUESTION PAPER 1

Chapter 14: Waves

Maximum Marks: 40 | Suggested time: 60 minutes

Instructions: Answer all questions. Show calculation steps and units. Original practice paper, not an official examination.

Section A 1-mark questions - 8 x 1 = 8 marks

1. Transverse oscillation relative to wave travel?
2. Sound in air type?
3. Wave speed relation?
4. Wave number k?
5. Superposition principle?
6. Reflection at fixed end?
7. Beat frequency?
8. What is amplitude?

Section B 2-mark questions - 6 x 2 = 12 marks

9. f=50Hz, lambda=2m: speed?
10. v=340m/s, f=170Hz: wavelength?
11. 256Hz and 260Hz beats?
12. Wavelength halves at fixed speed: frequency?
13. Reflection at free end: inversion?
14. Wave y=A sin(kx-omegat) travels which way?

Section C 3-mark questions - 4 x 3 = 12 marks

15. v=24m/s, lambda=.8m: f and period?
16. Displacements +3 cm and -1 cm overlap: resultant?
17. 300Hz and 304Hz: beat frequency and period?
18. How do waves transfer energy without permanent medium transport?

Section D 4-mark questions - 2 x 4 = 8 marks

19. Wave travels 12m in .4s with f=5Hz: speed/wavelength?
20. Explain beats for 440Hz and 442Hz.